

A Guide to Statistical Network Modeling

Session 2.1: Temporal ERGMs for Network Change

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instats Seminar

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Companion computations and reproducibility

- **Shared resources:** [repository](#) and [companion app](#); see Session 1.1 for the common reproducibility workflow.
- Open **Session 2.1** and choose one of **five visible public temporal-network workflows**; each exposes its waves, joint risk sets, transition counts, stated formula, conditional simulations, and downloadable audit record.
- **Computational boundary:** the app fits a first-order *lag-only* temporal ERGM whose included statistics depend on the preceding observed network; it does not silently fit a separable formation–dissolution model.
- **This session:** longitudinal design, conditional transition distributions, temporal dependence, risk sets, estimation distinctions, and transition-specific simulation checks.

Learning objectives

By the end of this session, we will be able to:

- define a **discrete-time temporal exponential-family random graph model** (TERGM) as a conditional distribution for an observed network transition;
- distinguish **persistent non-ties**, **formations**, **dissolutions**, and **persistent ties** on a declared risk set;
- formulate first-order persistence, delayed reciprocity, and lagged two-path effects without confusing them with contemporaneous structure;
- state the observation, support, homogeneity, and temporal-alignment assumptions required for a defensible transition model; and
- audit a fitted transition model through estimation diagnostics and conditional simulations rather than a coefficient table alone.

Session route

- ➊ **Design:** repeated networks, observation times, actor composition, risk sets, and covariate timing.
- ➋ **Model:** conditional transition probability, Markov order, temporal statistics, and parameter homogeneity.
- ➌ **Interpretation:** persistence, delayed effects, current structure, and the limits of temporal ordering.
- ➍ **Estimation and audit:** conditional likelihood, Monte Carlo likelihood, pseudo-likelihood, transition replication, and simulations.
- ➎ **No-code practice:** five public longitudinal networks and a constrained repeated-network BYOD workflow.

What changes when a network is observed repeatedly?

- A static ERGM assigns a probability distribution to **one graph-valued outcome** on a declared support.
- A TERGM assigns a conditional probability distribution to the next observed network **given network history** and pre-transition information.
- The response at time t is the whole network Y^t , not a collection of independently observed dyads.
- Temporal data add a second dependence dimension: a current tie can be associated with earlier ties while current ties may also be structurally interdependent.
- **Key distinction:** observing several snapshots does not itself determine the relevant time scale, risk set, or dynamic estimand.

Longitudinal network data

- Let $Y^1, \dots, Y^T$ denote networks observed at ordered discrete times $t = 1, \dots, T$.
- Let V^t be the actors present at time t and let D_t be the **admissible dyad set** at that time.
- A binary entry $Y_{ij}^t = 1$ records a tie under the stated measurement rule; $Y_{ij}^t = 0$ is a non-tie only when $(i, j) \in D_t$ is observed.
- The network boundary, node identifiers, directionality, mode structure, missingness, and meaning of a tie must be comparable enough to define a transition.
- Repeated relation layers, cumulative networks, and a single weighted aggregate are **not** automatically longitudinal network observations.

The support can change with time

- The support at wave t is $\mathcal{Y}_t$, the set of networks allowed by V^t , D_t , structural zeros, and any design constraints.
- Actor entry and exit can change V^t ; unavailable dyads then leave the risk set rather than becoming observed zeros.
- A bipartite network has within-mode structural zeros at every wave; these are not absent relations that a temporal model may form.
- Fixed-outdegree nomination designs constrain possible current networks differently from an unconstrained Bernoulli tie process.
- **Support consistency** is a modeling decision: it must be documented before calculating persistence or transition likelihoods.

First-order TERGM transition distribution

For a binary network with pre-transition covariates X^t ,

$$\mathbb{P}_\theta(Y^t = y^t \mid Y^{t-1} = y^{t-1}, X^t) = \frac{\exp\{\theta^\top g(y^t, y^{t-1}, X^t)\}}{\kappa_t(\theta; y^{t-1}, X^t)}.$$

- $g(\cdot)$ is a vector of **transition statistics** that may involve the current network, the preceding network, and pre-transition covariates.
- θ is a parameter vector; its components are conditional associations under the complete stated transition model.
- The probability is defined only on the admissible support $\mathcal{Y}_t$.

The transition normalizing constant

$$\kappa_t(\theta; y^{t-1}, X^t) = \sum_{z \in \mathcal{Y}_t} \exp\{\theta^\top g(z, y^{t-1}, X^t)\}.$$

- The sum ranges over all networks z that are allowed at the current wave, given its support restrictions.
- It depends on the prior observed network and may vary across transitions even when parameters are homogeneous.
- With contemporaneous structural dependence, evaluating κ_t is generally computationally intractable for realistic networks.
- This intractability motivates Monte Carlo likelihood methods; it does *not* license replacing unavailable dyads with zeros.

Conditional likelihood across observed transitions

Under a first-order Markov specification, conditional on Y^1 ,

$$L(\theta \mid y^1, \dots, y^T, X) = \prod_{t=2}^T \mathbb{P}_{\theta} \left(Y^t = y^t \mid Y^{t-1} = y^{t-1}, X^t \right).$$

- Under the first-order Markov specification, each conditional factor depends on the immediately preceding observed network rather than the full earlier history.
- It does **not** say that transitions are independent unconditional replicates: adjacent transitions share an observed network state.
- If the same θ is used at each transition, the model imposes **parameter homogeneity** across the selected observation intervals.
- Period indicators or interactions can relax homogeneity, but few observed transitions sharply limit their identification.

Markov order is a modeling restriction

- A first-order TERGM conditions on Y^{t-1} ; this does not prove that earlier history is irrelevant in the social process.
- An order- K model conditions on $Y^{t-1}, \dots, Y^{t-K}$ and contributes $T - K$ transitions to the conditional likelihood.
- Higher-order lags are useful only when the observation interval and substantive theory make them meaningful.
- Consecutive snapshots are often highly correlated, so adjacent lagged statistics can be nearly collinear.
- **Discipline:** choose lag order from theory and time scale, not unrestricted search over many available histories.

Homogeneous and heterogeneous transitions

- A **homogeneous TERGM** uses the same parameter vector θ for every modeled transition.
- This means that the conditional association represented by a statistic is assumed constant over the included observation intervals.
- A period-specific parameter or period-by-statistic interaction permits a stated form of temporal heterogeneity.
- With only a few transitions, an elaborate heterogeneous model can be weakly identified even when the network contains many dyads.
- Apparent temporal change may instead arise from changing risk sets, measurement protocols, missingness, or observation intervals.

Observation intervals define the estimand

- A monthly and a yearly network panel can encode different effective processes, even with the same actors and relation.
- At short intervals, most dyads may be unchanged and a stability effect can dominate every other term.
- At long intervals, formation and dissolution may both occur between observations; the endpoint transition loses their order and duration.
- A coefficient is therefore conditional on the stated **observation schedule**, not an automatically portable continuous-time rate.
- Unequal intervals should be represented, restricted, or justified; silently treating them as equal changes the model interpretation.

Transition-specific dyadic risk sets

- For a transition $t - 1 \rightarrow t$, let $D_{t-1,t} = D_{t-1} \cap D_t$ be the dyads observed and admissible at both endpoints.
- Only dyads in $D_{t-1,t}$ can be classified as formations, dissolutions, or persistent ties/non-ties.
- A person absent at one endpoint does not generate a tie dissolution with every other person.
- Incidental missingness is not evidence of a non-tie; structural unavailability is not missing-at-random zero data.
- In Session 2.1, every computation displays the joint risk-set size before fitting a transition model.

Actor composition is part of the model

- **Actor turnover** occurs when nodes enter, leave, or are not observable at a wave.
- A changing population can create apparent tie turnover if actor absence is incorrectly coded as a zero relation.
- A balanced continuing-actor subset can simplify a model, but it changes the target population and may discard relevant transition information.
- Alternatively, a transition-specific risk set preserves observed actors while excluding unavailable dyads.
- The chosen approach must be reported along with the network boundary and the interpretation of the resulting parameters.

Missingness is neither a tie nor a non-tie

- A response entry may be missing because an actor did not respond, a dyad was not measured, or a record was unavailable.
- A structural zero is an impossible dyad under the design; a missing value is an unobserved outcome; an observed zero is a measured non-tie.
- These three states have different implications for $\mathcal{Y}_t$, D_t , and the transition counts.
- Recoding all non-one values to zero inflates non-tie stability and can reverse a substantive temporal conclusion.
- The app requires an explicit at-risk-dyad table whenever node presence alone does not identify observed dyads.

Time-varying covariates require temporal alignment

- Let X^t denote covariate information available for modeling the transition $t - 1 \rightarrow t$; its measurement time must be stated explicitly.
- A sender, receiver, dyadic, or interaction effect must specify whether it uses X^{t-1} , X^t , or a documented interval summary.
- Using a covariate measured after a transition as a predictor creates a temporal-ordering problem.
- If an actor or covariate is unavailable at a wave, the affected transition contribution must be handled explicitly.
- Temporal precedence narrows some interpretations, but it does not by itself eliminate confounding or establish causality.

Four dyadic transition counts

For $(i, j) \in D_{t-1, t}$, define

$$N_{ab}^{(t)} = \sum_{(i, j) \in D_{t-1, t}} \mathbf{1}\{y_{ij}^{t-1} = a, y_{ij}^t = b\}, \quad a, b \in \{0, 1\}.$$

- $N_{00}^{(t)}$: persistent non-ties; $N_{01}^{(t)}$: formations.
- $N_{10}^{(t)}$: dissolutions; $N_{11}^{(t)}$: persistent ties.
- These are descriptive transition summaries on the declared joint risk set, before imposing a parametric TERGM.

Overall stability is not tie persistence

- **Overall dyadic stability** is $N_{00}^{(t)} + N_{11}^{(t)}$; it counts unchanged zeros and unchanged ones.
- **Tie persistence** is $N_{11}^{(t)}$ or, as a rate, $N_{11}^{(t)} / \{N_{10}^{(t)} + N_{11}^{(t)}\}$.
- In sparse networks, $N_{00}^{(t)}$ can dominate stability even when many existing ties dissolve.
- Therefore a high stability statistic is not evidence that existing relationships are durable.
- Separate reporting of all four counts prevents an undirected claim of “network persistence” from concealing this distinction.

Three sources of temporal structure

- **Contemporaneous structure:** a statistic involving only Y^t , such as current density or current closure.
- **Memory:** a statistic relating Y^t to Y^{t-1} , such as a same-dyad lagged tie.
- **History interaction:** a current tie associated with a prior reverse tie, prior two-path, or pre-transition covariate.
- These sources can coexist in a general TERGM, but their meanings and computational demands differ.
- A lagged term does not automatically account for every inherited feature of the preceding network.

Lagged same-dyad persistence

A common first-order statistic is

$$g_{\text{persist}}(y^t, y^{t-1}) = \sum_{(i,j) \in D_{t-1,t}} y_{ij}^t y_{ij}^{t-1}.$$

- It counts current ties that were ties at the preceding observed wave.
- Conditional on the other model terms, a positive coefficient increases the conditional log-odds of a current tie when the same prior tie was present.
- It is not a duration model: unobserved within-interval changes remain latent.
- A large persistence effect can absorb inertia that would otherwise be incorrectly attributed to current degree or closure structure.

Delayed reciprocity in a directed network

For a directed relation, a delayed-reciprocity statistic can be

$$g_{\text{drecip}}(y^t, y^{t-1}) = \sum_{(i,j) \in D_{t-1,t}^{\leftrightarrow}} y_{ij}^t y_{ji}^{t-1},$$

where $D_{t-1,t}^{\leftrightarrow}$ contains dyads for which the current $i \rightarrow j$ tie and prior reverse $j \rightarrow i$ tie are both observed and admissible.

- It records whether a prior incoming tie $j \rightarrow i$ is associated with a current outgoing tie $i \rightarrow j$.
- It is distinct from **contemporaneous mutuality**, which involves $y_{ij}^t y_{ji}^t$ at the same wave.
- It is undefined as a reciprocity mechanism for an undirected response network.
- A positive conditional association need not be a behavioral reply: stable dyadic propensities and unmeasured common causes remain possible.

Lagged two-path exposure

One possible prior-network exposure is

$$g_{2\text{path}}(y^t, y^{t-1}) = \sum_{(i,j) \in D_{t-1,t}} y_{ij}^t \sum_{k \neq i,j} y_{ik}^{t-1} y_{kj}^{t-1},$$

- The inner sum counts directed two-path opportunities from i to j in the preceding network and is restricted to observed, admissible prior-wave two-paths.
- It asks whether a current tie is conditionally associated with a prior two-path, not whether a current triangle is present.
- For an undirected network, the analogous prior shared-neighbor count is symmetric.
- This statistic should be included only when its one-wave lag and relation direction have a substantive interpretation.

Do not conflate current closure with inherited closure

- A current triangle statistic measures a property of Y^t and creates contemporaneous dependence among candidate ties.
- A lagged two-path statistic uses Y^{t-1} as fixed history when predicting Y^t .
- If a network is highly persistent, much current closure may be inherited from prior ties even when no new closure mechanism operates.
- Including both terms can be scientifically justified, but their collinearity and separate interpretations require diagnostic attention.
- Session 2.1 computes only lagged two-path exposure; it does not present a contemporaneous closure TERGM as if its normalizing constant were trivial.

The current-wave edges baseline

- An **edges** statistic counts current ties on the current transition support.
- Conditional on the other terms, its coefficient is a baseline log-odds contribution for a current tie with all included predictors equal to zero.
- It is not a universal density-control remedy: a constrained nomination design can make edge count constant, while a changing risk set changes the number of possible dyads.
- In an undirected sparse treaty panel, the baseline may be strongly negative; a memory effect is interpreted relative to that baseline.
- Baseline terms should be retained or omitted by support logic, not merely because software accepts them.

Temporal ordering does not create causal identification

- Conditioning on a prior network establishes that the modeled history precedes the current observed network.
- It does not rule out latent actor propensities, unmeasured dyadic opportunities, selection, common shocks, or measurement changes.
- A TERGM coefficient is therefore a **conditional association in a transition distribution** under the declared model and support.
- Causal claims require a separate design, treatment definition, interference assumptions, and identification argument.
- Avoid replacing “associated with” by “causes” merely because a predictor is lagged.

Exact conditional likelihood for a restricted lag-only model

- Suppose every fitted statistic is a function of Y^{t-1} and one candidate current dyad Y_{ij}^t , with no contemporaneous dependence among current dyads.
- Conditional on the observed history and risk set, the transition distribution then factorizes into logistic dyad probabilities.
- The resulting conditional logistic likelihood is **exact for that restricted TERGM subclass**, not a pseudo-likelihood approximation.
- Session 2.1 implements this transparent subclass for public data with limited transitions and reports its boundary in every computation record.
- It does not claim to estimate a general TERGM with current triangles, stars, or other coupled current-wave statistics.

Monte Carlo maximum likelihood for general TERGMs

- With contemporaneous dependence, the transition normalizing constants are intractable; as in static ERGMs, MCMLE approximates likelihood-based estimating equations using simulated networks.
- Its diagnostics include Monte Carlo error, MCMC mixing, convergence of the likelihood optimization, warnings, and repeated-fit stability.
- A completed optimization routine is not itself evidence that the chain explored the relevant transition distribution.
- General TERGM inference also requires enough transition information to support the selected history and structural terms.

Pseudo-likelihood is not Monte Carlo likelihood

- **As in static ERGMs**, pseudolikelihood multiplies individual-tie full conditionals whereas MCMLE targets the joint transition likelihood; with current-wave dependence their estimates and uncertainty statements can differ.
- Neither approach cures an undefined risk set, a weak observation design, or an unsupported actor-composition assumption.
- Reports should name the estimation method rather than using “TERGM estimation” as an undifferentiated label.

Temporal replication is not the number of dyads

- A large network may contain many dyads but only a few observed transitions.
- The amount of information about history effects depends on variation across at-risk dyads and transitions, together with the specified dependence structure; the number of observed transitions alone is not an effective sample size.
- A single Fall-to-Spring transition can estimate a restricted conditional model when sufficient dyadic variation exists, but it cannot identify temporal heterogeneity or support transition-block resampling.
- Resampling individual dyads treats them as independent temporal replications and is generally inappropriate for a network transition model.
- Where enough transitions exist, a **whole-transition bootstrap** is a limited sensitivity calculation, not a universal asymptotic guarantee.

Numerical and substantive transition audit

- Check that each selected term varies on the joint risk set and inspect large coefficients, extreme fitted probabilities, and ill-conditioned information matrices.
- Report the observed N_{00} , N_{01} , N_{10} , and N_{11} counts for every modeled interval.
- Simulate Y^t conditional on each observed Y^{t-1} and compare observed current tie totals, formations, dissolutions, persistence, and substantively relevant structural features with their simulated distributions.
- Inspect whether an apparent lack of fit is concentrated in particular periods, risk-set changes, or low-information intervals.
- A successful numerical fit is one component of an audit, not a proof that the chosen transition statistics are scientifically adequate.

A defensible Session 2.1 workflow

- 1 Define the repeated response relation, observation dates, network boundary, and all structural restrictions.
- 2 Construct wave-specific node presence and joint at-risk dyads; preserve missing and unavailable states.
- 3 Tabulate four transition counts before selecting a TERGM statistic.
- 4 State a parsimonious first-order conditional model linked to theory and the observation interval.
- 5 Fit with recorded settings, inspect numerical flags, and compare observed transition summaries with conditional simulations.
- 6 Report what the data cannot identify, including within-interval ordering, causal effects, and separate formation/dissolution processes.

Worked example: Knecht classroom friendship

- **Data:** four directed binary good-friend nomination waves in a Dutch classroom; one pupil leaves after wave 2.
- **Risk-set rule:** source codes for missingness and structural nonmembership are excluded; they are never converted into non-ties.
- **Stated model:** current edges, lagged same-dyad persistence, and delayed reciprocity on dyads observed at both adjacent waves.
- **Computation:** three transition tables, no transition-block bootstrap, and one-step simulations conditional on each previous nomination network.
- **Limit:** three transitions provide a compact teaching panel, not a basis for complex lags, influence claims, or continuous-time conclusions.

Worked example: Sampson repeated liking nominations

- **Data:** three directed positive-affect/liking panels among 18 monk-novitiates, distributed as separate public network objects.
- **Stated model:** edges, lagged persistence, delayed reciprocity, and prior directed two-path exposure.
- **Temporal distinction:** the two-path predictor is computed from the preceding panel, whereas a current transitive-triad term would be a different contemporaneous specification.
- **Computation:** two conditional transitions with simulated current tie totals, formations, and persistence.
- **Limit:** two transitions cannot disentangle a stable dyadic propensity from every possible social response mechanism.

Worked example: Coleman Fall-to-Spring nominations

- **Data:** two directed binary sociometric nomination networks among 73 high-school boys, recorded in Fall 1957 and Spring 1958.
- **Stated model:** a single conditional Fall-to-Spring transition with edges, persistence, and delayed reciprocity.
- **Support assumption:** the public array supplies a common roster but no wave-specific presence indicator; fixed risk is stated rather than inferred.
- **Computation:** descriptive transition counts and conditional simulations; the app withholds a transition-block bootstrap.
- **Limit:** one transition cannot estimate a temporal trend, period heterogeneity, lag-two memory, or robust temporal uncertainty.

Worked example: COW annual formal alliances

- **Data:** eleven annual undirected binary alliance networks, 1960–1970, built from active COW treaty dyads and annual state-system membership.
- **Response:** an edge indicates at least one active formal alliance for an unordered pair; it is not treaty multiplicity, military cooperation, or defense-only commitment.
- **Risk-set rule:** state nonmembership is structurally unavailable, so pre-entry and post-exit dyads are excluded rather than coded as non-alliance.
- **Stated model:** edges, lagged alliance persistence, and prior shared-partner exposure; ten annual transitions permit a cautious block bootstrap.
- **Limit:** annual coding obscures within-year ordering and long treaty spells can generate strong persistence without a causal dyadic response.

Worked example: Windsurfers daily interactions

- **Data:** repeated undirected binary daily observed-interaction panels at southern California beaches, with substantial attendance turnover.
- **Risk-set rule:** only dyads jointly present on adjacent days are modeled; absence from the beach is not an observed non-tie.
- **Time-gap rule:** the documented missing daily panel is retained as a break; no artificial one-step transition is formed across it.
- **Stated model:** edges, lagged interaction persistence, and prior shared-neighbor exposure, followed by transition-block simulations and bootstrap checks.
- **Limit:** daily interaction presence is not a durable friendship outcome, and small joint risk sets constrain detailed structural claims.

Why the Newcomb ranking data are not a worked binary TERGM

- **Data:** repeated complete preference rankings among fraternity members, rather than unconstrained binary tie observations.
- A documented top- k dichotomization fixes each respondent's outgoing selections and changes the support to a constrained choice process.
- Treating the transformed zeros as ordinary absent friendship ties would misrepresent both measurement and outdegree constraint.
- A rank-order or constrained choice model is more faithful to the original response; a binary TERGM requires a separately justified transformation and support.
- **Principle:** public availability does not make a data set method-appropriate; exclusion can be a correct modeling decision.

No-code app navigation

- ➊ Open the **Session 2.1** page of the Streamlit app hosted on Render.
- ➋ Choose one of the five visible public temporal networks; read its response boundary, support note, method appropriateness, and interpretation limit.
- ➌ Inspect the wave profile and four-count transition table before running a model.
- ➍ Run the stated first-order lag-only TERGM with a recorded seed, conditional-simulation count, and, where appropriate, transition-block bootstrap setting.
- ➎ Review coefficients, numerical flags, transition tables, conditional simulation envelopes, and the downloadable Markdown computation record.

BYOD: repeated-network upload contract

- **Node-presence table:** wave and id; an optional `transition_block` marks a break after a missing panel.
- **Temporal edge table:** wave, source, and target for observed binary ties.
- **At-risk-dyad table:** required whenever present actors do not imply that every within-wave dyad was observed.
- **Declare explicitly:** directionality, wave spacing, actor boundary, structural zeros, missingness, and whether adjacent observed waves define a meaningful one-step interval.
- **Not accepted silently:** valued/count ties, rank-order outcomes, multiplex layers, unknown support, actor absence as a zero tie, or a missing wave bridged as a transition.

A reporting checklist for Session 2.1

- **Design:** relation meaning, observation dates, interval comparability, actors, network boundary, directionality, and modes.
- **Support:** wave-specific presence, structural zeros, missingness rule, joint risk-set construction, and excluded transitions.
- **Specification:** Markov order, homogeneous/heterogeneous parameters, transition statistics, lag convention, and covariate timing.
- **Estimation:** method, software version, seed, simulation settings, numerical diagnostics, and treatment of limited transition replication.
- **Interpretation:** conditional association, conditional simulations, generalization regime, and explicit rejection of unsupported causal or continuous-time claims.

Session 2.1 synthesis

- A TERGM models a conditional distribution for observed network transitions; it is not simply a static ERGM repeated at several dates.
- Risk sets, actor composition, missingness, and observation intervals define the data to which transition statistics apply.
- Overall stability, tie persistence, formation, and dissolution are distinct descriptive summaries even before a model is fit.
- Lagged statistics encode selected temporal associations; they do not automatically establish causal effects or recover unobserved event histories.
- Next: **separable TERGMs**, which define distinct formation and persistence/dissolution components on different dyadic risk sets.

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